Why use multivariate data analysis?

A Comparison between Multiple Univariate Confidence Intervals and Multivariate Confidence Regions

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1 Introduction

Why do we use multivariate data analysis? When conducting inference through hypothesis testing, we require a quantifiable degree of uncertainty when dealing with sample data. The traditional way of doing this is through confidence intervals. Confidence intervals generate a range of values with a level of certainty to contain a true value of a population parameter from sample data. However, these intervals are only bounding one dimension at once. It is established that real world data is typically multifaceted, complex, and can't easily be oversimplified to any one factor.

The new problem becomes so: **How can we generate a "confidence interval" for multiple population parameters?** The naive way of solving this issue is to just generate multiple confidence intervals for each variable. However, this approach arises some concerns:

1. Each confidence interval is computed from its respective variable only, which forms a rectangle. This approach ignores correlation between variables, which would not offer guarantees in multiple dimensions when variables have relationships.
2. Joint coverage is not controlled. At $p = 2$ intervals with 95% confidence, we get a joint coverage of $.95^2 = .9025$ under independence, at a family-wise error rate of 9.75%. At $p = 5$, joint coverage drops to $.95^5 \approx .7738$ at a family-wise error-rate of 22.6%. The coverage drops and family-wise error rate increases, needing multiple testing correction.

In this report, we introduce the idea of the confidence region and how these issues are addressed. We show that a confidence region (ellipsoid) can offer calibrated simultaneous coverage of multiple parameters without simultaneous testing correction. We perform a simulation of bivariate multivariate normal data and compare the coverage of multiple confidence intervals to a confidence region. We show that a confidence region maintains coverage on average regardless of correlation and visualize the overarching geometric advantage that a confidence region has

over multiple confidence intervals. Unlike univariate confidence intervals, the confidence region generates a shape that incorporates correlation between variables. The limitation of confidence regions is that in higher dimensional settings, the region becomes difficult to describe and use compared to multiple univariate confidence intervals.

Data (Section 2) introduces how the simulated data is generated. Methods (Section 3) presents the mathematical formulations behind confidence intervals and regions, as well as how each method will be compared. Results (Section 4) shows the findings for the comparisons between our methods, and Discussion (Section 5) deliberates the significance of our findings in the simulation.

2 Data

We generate the data matrix $\mathbf{X}$ using the bivariate normal distribution choosing the number of observations n = 50, mean $\boldsymbol{\mu} = \begin{bmatrix} 0 & 0 \end{bmatrix}^T$ and a variable covariance and diagonal variance of ones. This simplifies our example, as the off-diagonal simplifies to our correlation ρ and all assumptions required of standard confidence intervals and regions are satisfied. We define correlation to be

$$\rho = \frac{Cov(X_1, X_2)}{\sigma_{X_1}\sigma_{X_2}}$$

The result is a bivariate normal random vector:

$$\mathbf{X} \sim N_2(\boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \boldsymbol{\Sigma} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix})$$

For a high correlation of $\rho = .9$, a sample of $n = 50$ observations would look like

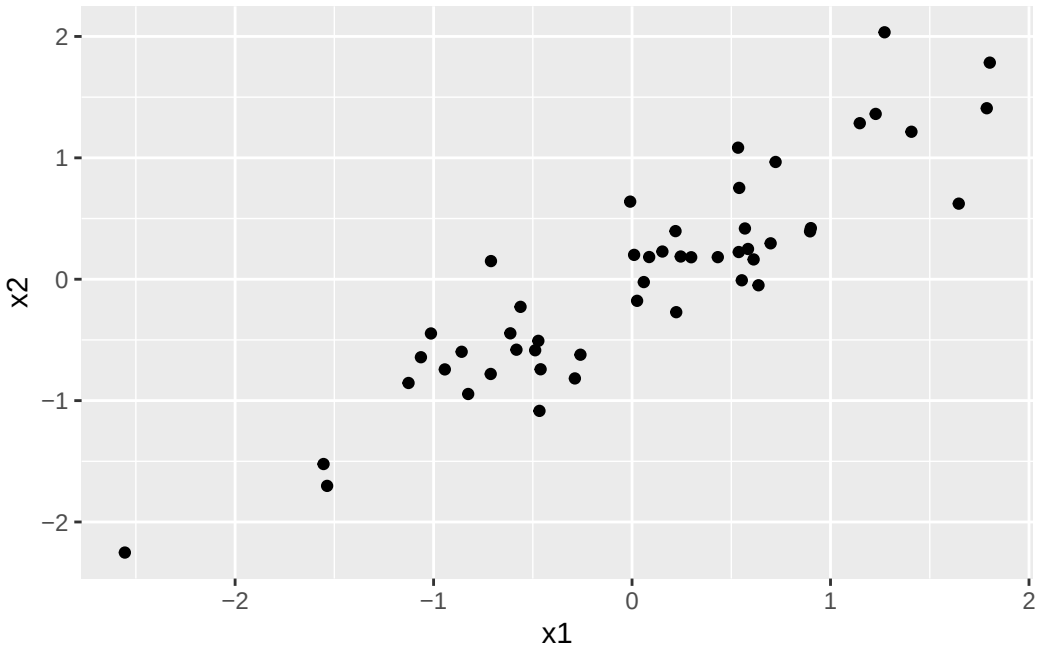


Figure 1: Scatterplot of x1 against x2 in High Correlated Environment

For a low correlation of $\rho = .05$, a sample of $n = 50$ observations would look like

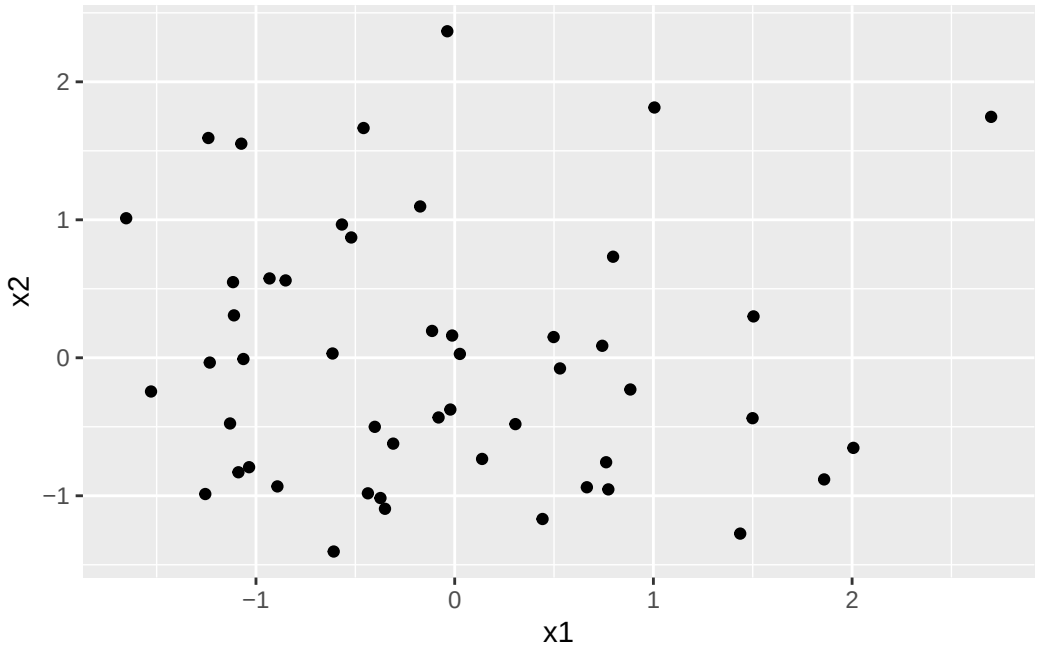


Figure 2: Scatterplot of x1 against x2 in Low Correlated Environment

3 Methods

Recall the univariate (100(1-α))% t-based confidence interval for a population mean with significance level α, sample size n, sample standard deviation s, and $t_{n-1}(\alpha/2)$ is the critical value of the t distribution with df=n-1:

$$\bar{x} \pm t_{n-1}(\alpha/2)\frac{s}{\sqrt{n}}$$

This confidence interval rejects the set of all values μ_0 outside of its bounds and fails to reject the set of all values μ_0 inside its bounds at level α, corresponding to the null hypotheses

$$H_0 : \mu = \mu_0$$

and alternative hypotheses

$$H_1 : \mu \neq \mu_0$$

A confidence region for the mean vector of a p-dimensional normal distribution is the ellipsoid containing all μ such that

$$n(\bar{\boldsymbol{x}} - \boldsymbol{\mu})^T \boldsymbol{S}^{-1}(\bar{\boldsymbol{x}} - \boldsymbol{\mu}) \leq \frac{p(n-1)}{(n-p)} F_{(p,n-p)}(\alpha)$$

where $\bar{\boldsymbol{x}}$ and $\mathbf{S}$ are the sample mean vector and sample covariance matrix, $\boldsymbol{\mu}$ is the population mean vector and $F_{(p,n-p)}(\alpha)$ is the upper α critical value of the F distribution with p and n-p df. This corresponds to the Hotelling's T^2 test with the null hypothesis

$$H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$$

and alternative hypothesis

$$H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$$

The region rejects the set of all values of $\boldsymbol{\mu}_0$ outside of its bounds and fails to reject all values of $\boldsymbol{\mu}_0$ inside of its bounds at level α.

We will begin by generating and comparing confidence intervals and confidence regions for both our lowly-correlated and highly-correlated data at a standard 95% confidence level. On bivariate data, the "joint" confidence intervals form a bounding rectangle, while the confidence region's geometry forms an ellipse. We then will continue with a high-volume simulation comparing the two methods as correlation increases. We expect that since the joint-confidence intervals are essentially performing two separate tests, the joint coverage will start at around $.95^2 = .9025$ and rise with ρ while the confidence region's coverage will stay the same.

We implement the analysis using R (R Core Team 2025). Multivariate normal data generated with mvtnorm (Genz and Bretz 2009). Plots done with ggplot2 (Wickham 2016). The T^2 based confidence region for a mean vector and univariate t-based confidence interval follows Johnson and Wichern (2007). Code template for confidence region and interval comparison generated by Dr. Gao.

4 Results

Figure 3 and Figure 4 show the 95% confidence region and intervals for our two observations, x1 and x2, projected onto one sample of our low and high correlated data. There are a couple things to note:

1. The bounding box generated by the confidence intervals representing the respective sample parameters overlap the area associated with ellipsoid confidence regions. The bounding box contains extra area in the form of "corners" while the ellipse contains extra area following the correlation of the data at the ends of each ellipse, extending past the box.
2. The confidence region generated with low-correlated data is more circular in nature than the high-correlated data, which is more elongated. Due to this elongation, there seems to be larger "corners" in the high-correlated data.

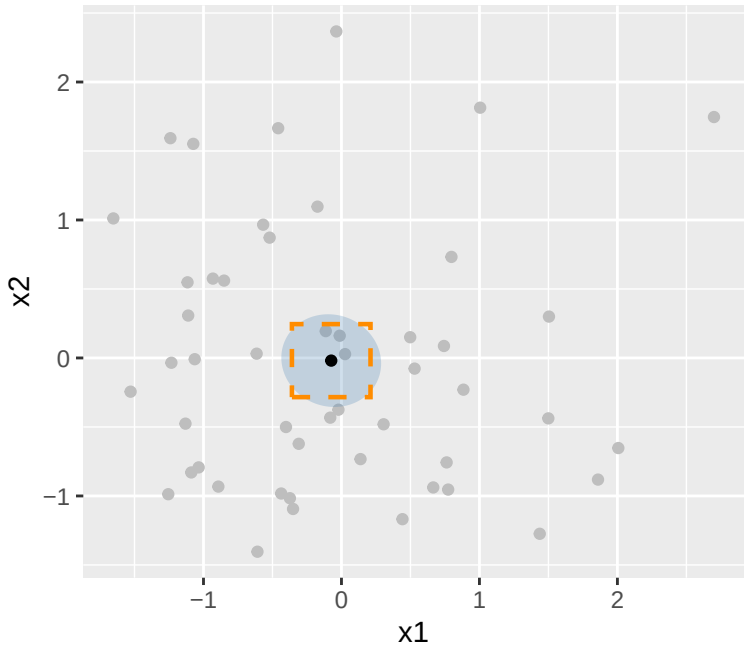


Figure 3: Multiple Confidence Intervals vs Regions on Bivariate normal data with low correlation

We generate the situation above with varying correlations at 50,000 simulations each to show how as the correlations change, the confidence intervals falls below 95% confidence while the confidence regions hold.

Figure 5 shows a simulation of 50,000 for 13 different ρ at size n = 50 samples of data, starting at 50,000 samples at $\rho = 0$ and 50,000 additional samples from 0 to .9 at steps of .1, plus at .95, .975, and .99 for 650,000 total samples at 95% confidence. As hypothesized, the confidence

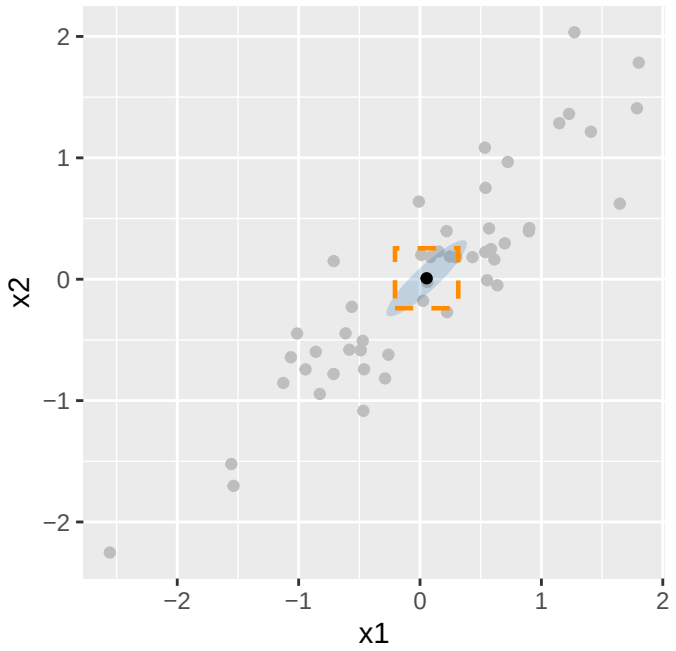


Figure 4: Multiple Confidence Intervals vs Regions on Bivariate normal data with high correlation

region maintains 95% coverage while the joint confidence interval "region" coverage slowly increases to around .95 as ρ approaches 1.

5 Discussion

From our simulation in Figure 5, the confidence interval approach at $\rho = 0$ for two confidence intervals had a joint coverage of about 90%, which rose to around 95% as ρ increased for $p = 2$. Univariate confidence intervals do not account for correlation, so when one variable is strongly correlated with another, variables tend to pass and fail the intervals together, and the probability that at least one interval fails drops. The multivariate confidence region stayed at around 95% coverage as ρ increased. In a real study, this is important because we do not know the true correlation between population variables. The actual joint coverage could be about 90 to 95%, and therefore the original 95% level cannot be fully trusted.

Coverage is only one part of the problem. In Figure 3 and Figure 4, the bounding box "corners" that the ellipse did not share were also due to the confidence intervals' failure to account for correlation within the data. The ellipse that the confidence region generated excludes those implausible corners by properly accounting for correlation, ruling them out. The ellipse also extends past the boxes in the direction of correlation and becomes more elongated as correlation increases. These differences are emphasized when visually comparing the area of the corners

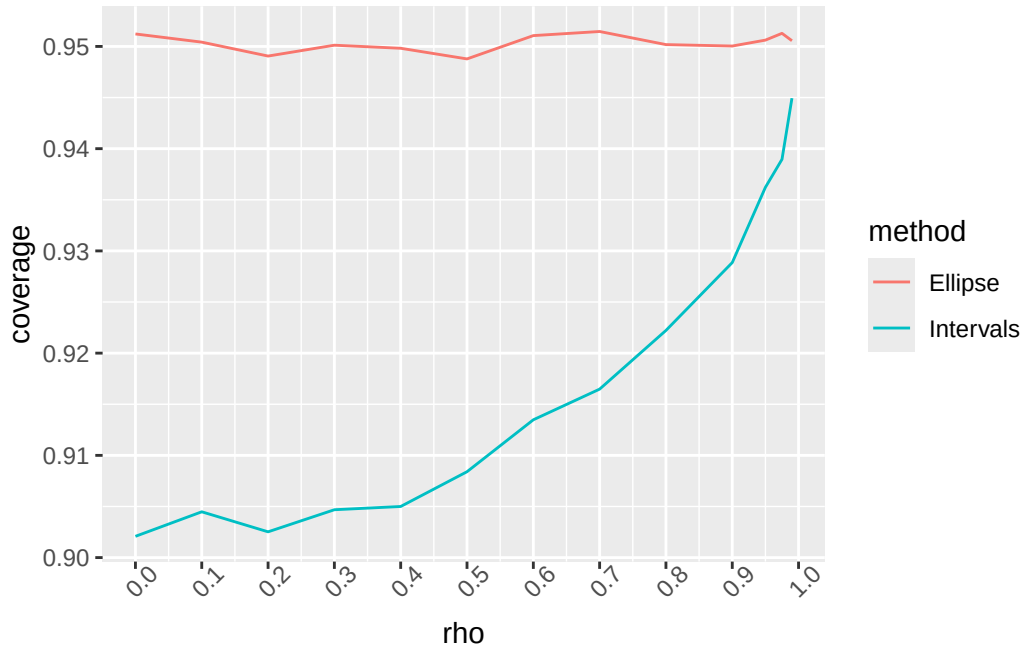


Figure 5: Evaluating the confidence region and interval performance on average for 50000 iterations and as rho increases for confidence level .95

and elongated ellipses in the low and high correlation examples. At high ρ, a region that accounts for correlation can achieve better coverage for less wasted area than a box.

While univariate methods fail to account for correlation, they are easier to interpret than confidence regions. Considering a simulation with 3 variables, the confidence "ellipse" turns into a confidence "ellipsoid" in 3D. At 4 and above variables, the ability to conceptualize the results diminishes.

Limitations and Extensions

We simulated only bivariate data with a set sample size (n = 50), equal variances, and only one confidence level for the sake of simplicity. One could extend this simulation to include variation in sample size, variance, and confidence level to get a complete picture of how correlation interacts with univariate confidence intervals. We also only measured coverage and visually compared area. Additional analysis could include computed area and power.

When using multiple univariate confidence intervals, the probability of at least one confidence interval missing inflates with the addition of each additional variable. Adding Bonferroni-corrected intervals would restore joint-coverage of at least 95% at every ρ. One could extend this simulation to show how the family-wise error rate of p univariate intervals grows with p in

a p-variate normal distribution. This extension could include an additional comparison with Bonferroni-corrected confidence intervals as the number of dimensions increases.

References

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